Effects of Browser Tab Orientation & Tab Load on Navigation Performance

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Abstract

Browser tabs support access to multiple webpages, but navigating large tab sets can make specific pages difficult to locate. This study investigates how tab orientation and tab load affect desktop navigation performance. A controlled 2×2 within-subjects experiment will involve 16 participants completing retrieval tasks using horizontal and vertical tabs under low and high loads. Condition order will be counterbalanced. Screen recordings will provide target-tab retrieval time, first-selection accuracy, incorrect-selection counts, and total tab switches. Perceived workload will be measured using the weighted NASA Task Load Index (NASA-TLX). Participants will rate six dimensions after each condition and complete one pairwise-comparison exercise after all conditions. The study will examine whether orientation effects change with tab load, providing evidence to inform browser interface design.

1 Introduction

Web browsers are widely used for academic, professional, and personal activities that require users to access and revisit information across multiple webpages. However, locating a specific webpage becomes increasingly difficult as the number of open browser tabs increases. Although tabs allow users to retain references and switch between ongoing tasks, accumulating too many tabs can reduce the space available for individual tabs, causing webpage titles to become truncated and making webpages harder to distinguish. As a result, users may spend more time searching for a particular webpage, select incorrect tabs, or repeatedly switch between tabs before finding the information they need. These difficulties make browser tab navigation an important human-computer interaction (HCI) concern because inefficient information retrieval can increase task completion time and the mental effort required to browse effectively. Previous research has documented this tension between the usefulness of tabs for retaining information and the navigation difficulties associated with accumulated tabs (Chang, Hahn, et al., 2021).

One interface characteristic that may influence the difficulty of locating a webpage is browser tab orientation. Horizontal tab bars, commonly positioned at the top of desktop browsers, arrange tabs in a row, whereas vertical tab layouts position them along the side of the window. These arrangements differ in how they use screen space and display webpage titles, potentially affecting how easily users can distinguish and identify a target tab. In horizontal layouts, increasing the number of open tabs may cause individual tabs to become narrower, reducing the visibility of their titles. Vertical layouts may provide more room for displaying webpage titles but also occupy space that could otherwise be used for webpage content. Consequently, neither orientation can be assumed to be more effective in every situation. The number of simultaneously open tabs may further influence navigation performance because users must search through a larger set of competing webpages as tab load increases. Understanding how these two factors work together is therefore important for determining which layout better supports target-tab retrieval under different browsing conditions.

Previous studies have examined navigation performance across different interface orientations and numbers of items. Backs et al. (1987), for example, compared horizontal and vertical menu formats across different menu sizes, finding differences in search time between orientations and longer search times for larger menus. Similarly, Warr and Chi (2013) compared horizontal page switching and vertical page scrolling in a mobile browsing context, examining selection performance and user experience across different numbers of pages. These studies provide a basis for investigating how interface arrangement and the number of available items may influence navigation. However, website menus, mobile page-switching interfaces, and desktop browser tab bars differ in their functions and spatial constraints. Their findings therefore do not directly establish how horizontal and vertical desktop browser tab layouts affect target-tab retrieval. Other studies have investigated browser tab-management systems that introduce organizational features to help users manage accumulated webpages. For example, Chang, Kim, et al. (2021) evaluated Tabs.do, a system that organizes tabs into task-based groups, while Rutishauser and Fritz (2026) investigated webpage management and retrieval using Gstell. Although these studies address difficulties associated with managing and rediscovering webpages, their approaches do not isolate the effect of tab orientation on retrieval performance.

These limitations indicate the need for a controlled comparison of horizontal and vertical desktop browser tab layouts under different tab-load conditions. In particular, it is important to examine whether the effectiveness of each orientation changes as the number of open tabs increases. A layout that supports efficient retrieval when only a few tabs are open may not provide the same advantage when users must distinguish among a larger number of webpages. Furthermore, retrieval time alone may not fully capture the usability of a tab layout. Users may locate a webpage in a similar amount of time across two layouts while experiencing different levels of difficulty or making different numbers of incorrect selections. Examining target-tab retrieval time, first-selection accuracy, and perceived workload, together with incorrect-selection counts and total tab switches as additional descriptive measures, can provide a more comprehensive understanding of how tab orientation and tab load affect navigation performance and user experience.

To address this research gap, the present study will investigate the effects of browser tab orientation and tab load on users' navigation performance during multi-tab browsing. Using a controlled, within-subjects experimental design, each participant will complete target-tab retrieval tasks under four conditions: horizontal tabs with low tab load, horizontal tabs with high tab load, vertical tabs with low tab load, and vertical tabs with high tab load. Screen recordings will be manually reviewed to measure target-tab retrieval time, first-selection accuracy, incorrect-selection counts, and total tab switches. Perceived workload will be measured using the weighted NASA Task Load Index (NASA-TLX). By examining the effects of orientation and tab load individually and in combination, the study aims to determine whether one orientation provides a retrieval advantage and whether this advantage varies with the number of open tabs. The findings may contribute to the evaluation and design of desktop browser interfaces by providing evidence on how tab orientation and tab load influence browsing efficiency and perceived workload.

The Related Work section reviews previous studies and establishes the research gap. The Preliminaries section defines the study variables and measurement rules, while the Methodology section presents the proposed experimental design, participants, materials, procedures, and planned data analysis. The Future Work section outlines the immediate next steps for preparing and conducting the study.

2 Related Work

2.1 Browser Tab Use, Clutter and Re-finding

Browser tabs support ongoing work by keeping webpages available for switching and revisiting. Using browsing logs and interviews, Dubroy and Balakrishnan (2010) found that participants valued tabs for convenient access and organization compared with multiple browser windows. Chang, Hahn, et al. (2021) further explained why users retain tabs, identifying their use as reminders, references, and a way to avoid searching for information again. Whereas Dubroy and Balakrishnan characterize how tabs support navigation, Chang et al. show why users may continue accumulating tabs even when doing so increases management demands.

Keeping these pages available can also introduce navigation difficulties. Chang, Hahn, et al. (2021) identified limited attention and screen space as pressures for closing tabs and reported that shrinking titles and icons can make open pages more difficult to identify. Ma et al. (2023) similarly associated tab accumulation with greater reported navigation clutter and a perceived excess of tabs and windows. Although both studies examine browsing clutter, Chang et al. focus on the competing pressures to retain or close tabs, whereas Ma et al. examine relationships among browsing habits, clutter, and coping strategies.

Together, these findings suggest that increasing tab load may affect navigation through two related mechanisms. First, users must search among a larger number of competing tab representations. Second, as available interface space becomes more constrained, the identifying information presented by each tab may become less visible. These effects may make target tabs more difficult to distinguish and re-find, potentially increasing retrieval time, incorrect selections, and perceived workload. This is particularly relevant when comparing horizontal and vertical tab layouts because the two orientations allocate screen space to tab labels differently. However, the studies above primarily describe browsing behavior and reported clutter rather than directly measuring target-tab retrieval under controlled orientation and load conditions. The present study therefore examines these factors experimentally.

2.2 Orientation and Navigation Performance

Studies of navigation layouts report different findings on search and selection performance. In a controlled menu-search task, Backs et al. (1987) found significantly shorter search times for vertical menus, whereas Magnussen et al. (2024) found no significant differences in average completion time or error rate between horizontal, vertical, and radial website menus. However, participants in Magnussen et al. preferred horizontal menus, supporting the need to examine preference alongside measured performance. Warr and Chi (2013) also found faster switching with vertically scrolling Cards than horizontally swiping Pages in a mobile interface. However, Cards displayed more pages at once and required fewer interactions, so its advantage cannot be attributed to orientation alone. These studies examine related navigation problems, but their different tasks and interfaces limit how directly their findings apply to desktop browser tabs.

The number of available items provides another connection between these studies. Backs et al. found that larger menus took longer to search, while Warr and Chi found lower perceived success and greater reported effort and frustration with larger page sets. Magnussen et al. kept menu size fixed and therefore did not test the effects of item count. Evidence about errors is also limited: Backs et al. did not statistically analyze accuracy, while Warr and Chi's larger page sets involved more trials, meaning that their higher total error counts do not necessarily indicate a higher error rate per selection. Neither Backs et al. nor Warr and Chi detected a significant interaction between layout and item or page count for time. These findings motivate examining horizontal and vertical desktop tabs under low and high tab loads, measuring target-tab retrieval time, first-selection accuracy, and perceived workload, and testing whether differences between orientations change with tab load. Incorrect-selection counts and total tab switches will provide additional descriptive measures of navigation activity.

2.3 Tab Management and Browser Design

Browser-management systems have explored how organization and visual presentation can support users' ongoing work. Rutishauser and Fritz (2026) and Chang, Kim, et al. (2021) both introduced methods for retaining webpages without requiring every page to remain continuously active, although their approaches differed. Gstell allowed users to temporarily shelve pages before organizing them into groups, whereas Tabs.do organized tabs into task bundles that could be saved and reopened together. Gstell users reported improved focus, while Tabs.do participants described benefits for task organization, resuming work, and confidently closing tabs. Lottridge et al. (2012) instead modified the visual prominence of work and non-work tabs and added timing information to support attention and time awareness. During the modified-browser phase,

participants switched tabs less often, kept fewer tabs open per day, and reported greater time awareness.

These studies demonstrate that browser-management interfaces can influence both browsing behavior and users' subjective experience. However, they also illustrate a methodological difficulty for studying tab layout: the evaluated systems changed several interface features at the same time. Shelving, grouping, visual prominence, and timing feedback provide different forms of assistance, making it difficult to determine how much of the reported benefit can be attributed to any single interface characteristic. Changes in overall tab counts or switching frequency also do not directly establish whether users can retrieve a particular target tab more quickly or accurately.

These limitations inform the design of the present study. Rather than evaluating a complete tab-management system, the experiment manipulates only tab orientation and tab load while keeping other organizational features constant. Automatic grouping, pinned tabs, tab-search functions, and other navigation aids will therefore not be used during scored trials. By holding these features constant, the study aims to attribute differences in retrieval performance and perceived workload more directly to the experimental variables rather than to additional organizational assistance.

2.4 Perceived Workload, Familiarity, and Learning Effects

Perceived workload can complement target-tab retrieval time and first-selection accuracy by capturing how demanding navigation feels to users. Warr and Chi (2013) used a simplified NASA-TLX covering perceived success, effort, and frustration, finding lower perceived success and greater effort and frustration as page count increased. These findings support examining users' reported experience alongside their observed performance. Hart and Staveland (1988) developed NASA-TLX as a multidimensional measure of subjective workload, providing an established instrument for this purpose. Its performance dimension concerns perceived success rather than recorded selection accuracy. This study therefore includes perceived workload alongside target-tab retrieval time and first-selection accuracy when comparing horizontal and vertical desktop tabs under low and high tab loads.

Interpreting these outcomes also requires attention to prior experience and learning. Magnussen et al. (2024) provided familiarization before testing each menu style, while Backs et al. (1987) excluded initial practice trials from analysis and balanced which orientation participants encountered first. Familiarization and practice help participants understand and operate the interfaces, whereas counterbalancing helps distribute condition-order effects across participants. However, Backs et al. suggested that participants might still have been improving despite practice, indicating that these procedures do not necessarily eliminate learning. Accordingly, this study will document prior experience with both tab orientations, provide practice, and counterbalance the four conditions to help account for these influences when interpreting performance and workload differences. Remaining differences in familiarity and learning should still be acknowledged.

2.5 Research Gap

Previous studies have documented frequent tab switching and reported difficulties associated with managing accumulated tabs. Other studies have compared orientation or switching interfaces across different item counts in menu and mobile settings. However, these tasks differ from retrieving desktop browser tabs, where title visibility, favicons, familiarity, and available screen space may influence performance. Tab-management systems such as Tabs.do and Gstell also introduce broader organizational features, so their evaluations do not isolate the contribution of orientation. Among the studies reviewed, no controlled comparison was identified between horizontal and vertical desktop browser tab layouts across specified tab loads. Through a controlled and counterbalanced experiment, this study will examine target-tab retrieval time, first-selection accuracy, and perceived workload, including whether differences between orientations change with tab load. Incorrect-selection counts and total tab switches will provide additional descriptive measures of navigation activity.

3 Preliminaries

3.1 Tab Orientation and Tab Load

Browser tab orientation refers to the arrangement of tabs within a single browser window. Horizontal tabs appear in a row along the top of the window, while vertical tabs appear in an expanded panel along its side. The vertical panel's width will remain fixed throughout each participant's vertical-tab conditions.

Tab load refers to the number of task-related webpages available for retrieval in the study window. The low-load condition will contain five task-related tabs, while the high-load condition will contain fifteen. In addition, every condition will include one standardized neutral starting tab that contains no task-relevant information and will never serve as a retrieval target. Participants will therefore see six total tabs in the low-load condition and sixteen total tabs in the high-load condition. Because the neutral tab is held constant across all conditions, the experimental tab-load manipulation concerns the five versus fifteen task-related webpages. Playing the auditory task prompt will not alter the number of open tabs.

3.2 Navigation Performance Measures

The primary navigation-performance outcomes will be target-tab retrieval time and first-selection accuracy. Incorrect-selection counts and total tab switches will be retained as additional descriptive measures of navigation activity. Each retrieval trial will ask participants to locate a predefined target tab from a prompt describing the information being sought. Each retrieval trial will begin with the standardized neutral starting tab active. The researchers will manually review screen recordings to identify tab activations and their video timestamps.

Target-tab retrieval time. Retrieval time will be measured in seconds from the standardized verbal start cue in the pre-recorded auditory prompt to the first recorded frame showing the correct target tab as active. For each trial, a pre-recorded prompt describing the retrieval task will be played and will conclude with the cue "Go." Participants will be instructed not to begin navigating until this cue is heard. The screen recording will include system audio or

microphone capture so that the start cue can be identified during manual coding. Retrieval time will be calculated as

$$T = t_{\text{target}} - t_{\text{start}},$$

where t_{start} is the timestamp of the "Go" cue and t_{target} is the timestamp at which the correct target tab first becomes active. Incorrect selections will not stop timing. This measure therefore represents the time required to locate and activate the target after the participant has heard the complete prompt.

First-selection accuracy. A selection will mean activating a different tab. The standardized neutral starting tab, which is already active at the beginning of each trial, will not count as a selection. Clicking the currently active tab will likewise not count as another selection. However, if a participant navigates away from the neutral starting tab and later returns to it, that activation will count as a selection and will be considered incorrect because the neutral tab will never serve as a retrieval target.

Each valid scored trial with an observable first selection will receive a score of 1 if that selection is the target and 0 if it is not. First-selection accuracy will be calculated separately for each participant and condition as the percentage of these trials with a correct first selection. Trials with no observable first selection will be reported separately and excluded from this calculation. The percentage of incorrect first selections will use the same denominator and equal 100% minus first-selection accuracy.

Incorrect-selection counts. Incorrect selections will be recorded as the number of non-target tab activations before the first correct target activation in completed trials. For incomplete trials, non-target activations will be counted until the trial ends and reported alongside the trial's completion status. Returning to a previously selected non-target tab will count as another incorrect selection. The count includes an incorrect first selection and any subsequent incorrect selections; it does not establish whether they were accidental. Researcher setup and reset actions outside the scored trial will not be counted.

Total tab switches. Total tab switches will count all activations during the trial, including the final correct selection when one occurs. For a completed trial, this total equals the number of incorrect selections plus one. It will therefore be retained as an additional descriptive measure of navigation activity, with its relationship to the incorrect-selection count made explicit.

3.3 Perceived Workload and Qualitative Feedback

Perceived workload refers to participants' assessment of the demands and effort involved in completing the tasks. The NASA Task Load Index (NASA-TLX), developed by Hart and Staveland (1988), measures six dimensions: mental demand, physical demand, temporal demand, perceived performance, effort, and frustration. This study will use the original weighted scoring procedure, combining six ratings with weights obtained from pairwise comparisons.

Participants will rate each dimension on a 0–100 scale in five-point increments after each condition. Higher values will indicate greater demand, effort, or frustration; the performance scale will run from good performance at 0 to poor performance at 100. After all four conditions, each participant will complete one set of 15 comparisons between pairs of dimensions, choosing which dimension contributed more to workload across the target-tab retrieval tasks. Because the four conditions share the same task structure, that participant's weights will be applied to all four conditions.

Each dimension's weight will equal the number of comparisons in which it is selected, ranging from zero to five, with all six weights summing to 15. For each condition, each rating will be multiplied by its corresponding weight. The six products will be added and divided by 15 to obtain one overall workload score between 0 and 100, with higher scores indicating greater perceived workload. The performance rating is a subjective assessment and is distinct from recorded first-selection accuracy.

After completing the six NASA-TLX ratings for each condition, participants may provide a brief neutral comment about the condition's overall difficulty. Detailed open-ended questions about navigation difficulties, strategies, and orientation preferences will be asked during the final post-study interview, after all four conditions and the NASA-TLX weighting exercise. The facilitator will document participants' verbal responses in written notes to provide qualitative context for the numerical results. No experience-related questions will be asked during timed retrieval trials.

4 Methodology

4.1 Methodology Overview

This study will use a controlled 2×2 within-subjects experiment to examine how browser tab orientation and tab load affect navigation performance and perceived workload. The planned sample will include 16 participants, each completing all four conditions: horizontal-low, horizontal-high, vertical-low, and vertical-high. Low and high tab-load conditions will contain five and fifteen task-related tabs, respectively. One additional neutral starting tab will remain open in every condition, resulting in six total visible tabs under low load and sixteen under high load. Participants will complete six target-tab retrieval trials per condition, resulting in 24 scored trials per participant. A Latin-square schedule will counterbalance condition order, and the assignment of four content sets to conditions will also be counterbalanced across participants.

Each session will include pre-study preparation, the four experimental conditions, and a post-study discussion. Before the scored trials, participants will provide informed consent, answer background questions, and practice using both tab orientations with separate practice webpages. Each participant will use the same device and settings throughout all four conditions. Screen recordings will be manually coded to obtain retrieval time, first-selection accuracy, incorrect-selection counts, and total tab switches. After each condition, participants will complete the six NASA-TLX ratings and may provide a brief neutral comment on overall difficulty. Detailed strategy and preference feedback will be collected during the final comparative interview. After all four conditions, they will complete one pairwise-comparison exercise to determine their workload weights before the final comparative interview and debriefing.

4.2 Research Questions and Hypotheses

This study is guided by the main research question: *How do browser tab orientation and tab load affect users' navigation performance*

Table 1: Latin-square condition orders and counterbalanced content-set assignments. Four participants are assigned to each order; each participant completes every condition and content set once.

Order group	Condition sequence	Content-set sequence
1	H-L → H-H → V-L → V-H	A → B → C → D
2	H-H → H-L → V-H → V-L	C → D → A → B
3	V-L → V-H → H-L → H-H	D → C → B → A
4	V-H → V-L → H-H → H-L	B → A → D → C

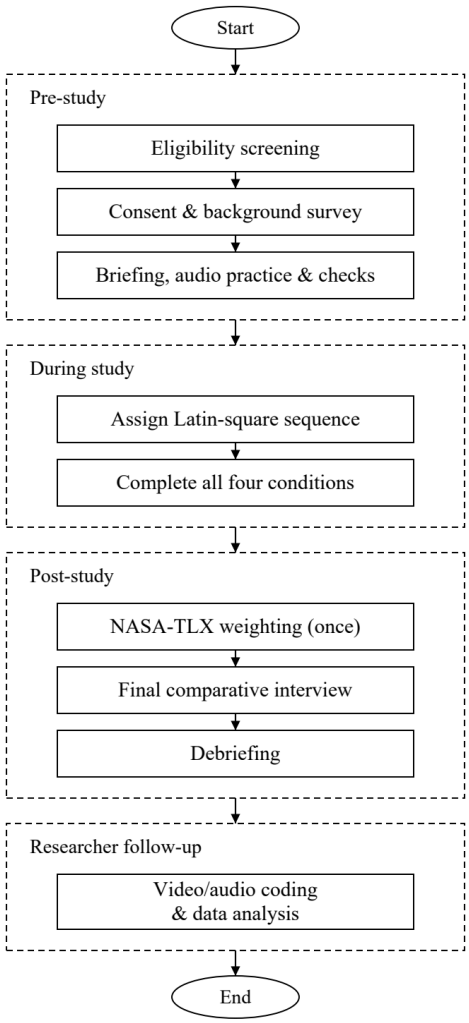


Figure 1: Overall study procedure. The experimental phase is expanded in Figures 2, 3, and 4; post-study weighting precedes the comparative interview, and video coding and data analysis follow the participant session.

during multi-tab browsing tasks? Navigation performance will be evaluated through target-tab retrieval time and first-selection accuracy. Perceived workload will be assessed as a complementary outcome using the weighted NASA-TLX score defined in the Preliminaries.

To address this question, RQ1 asks how these outcomes differ between horizontal and vertical tab orientations, on average across the two tab-load levels. RQ2 examines how the same outcomes differ between low and high tab loads, on average across the two orientations. RQ3 asks whether the difference between orientations changes depending on tab load. RQ4 explores the aspects of the conditions that participants describe as making retrieval easier or more difficult, together with their reported navigation strategies and orientation preferences.

The primary hypothesis concerns the interaction between tab orientation and tab load in RQ3. Supporting hypotheses concern the orientation main effect in RQ1 and the tab-load main effect in RQ2. Each hypothesis pair covers target-tab retrieval time, first-selection accuracy, and perceived workload.

Primary hypothesis: RQ3, interaction between tab orientation and tab load. $H_{0,3}$: The difference between horizontal and vertical tab orientations is the same under low and high tab loads for all three outcomes: target-tab retrieval time, probability of a correct first selection, and perceived workload measured using weighted NASA-TLX.

$H_{1,3}$: The difference between horizontal and vertical tab orientations changes between low and high tab loads for at least one of these three outcomes.

Supporting hypothesis: RQ1, effect of tab orientation. $H_{0,1}$: Averaged across low and high tab loads, horizontal and vertical tab orientations do not differ in target-tab retrieval time, probability of a correct first selection, or perceived workload measured using weighted NASA-TLX.

$H_{1,1}$: Averaged across low and high tab loads, horizontal and vertical tab orientations differ in at least one of these three outcomes.

Supporting hypothesis: RQ2, effect of tab load. $H_{0,2}$: Averaged across horizontal and vertical tab orientations, low and high tab loads do not differ in target-tab retrieval time, probability of a correct first selection, or perceived workload measured using weighted NASA-TLX.

$H_{1,2}$: Averaged across horizontal and vertical tab orientations, low and high tab loads differ in at least one of these three outcomes.

Each hypothesis is evaluated through outcome-specific tests, described under Data Analysis, and a null hypothesis is rejected if its effect is significant for at least one of the three outcomes. We expect vertical tabs to provide a greater advantage over horizontal tabs under high tab load than under low tab load, reflected in shorter retrieval times, higher first-selection accuracy, and lower

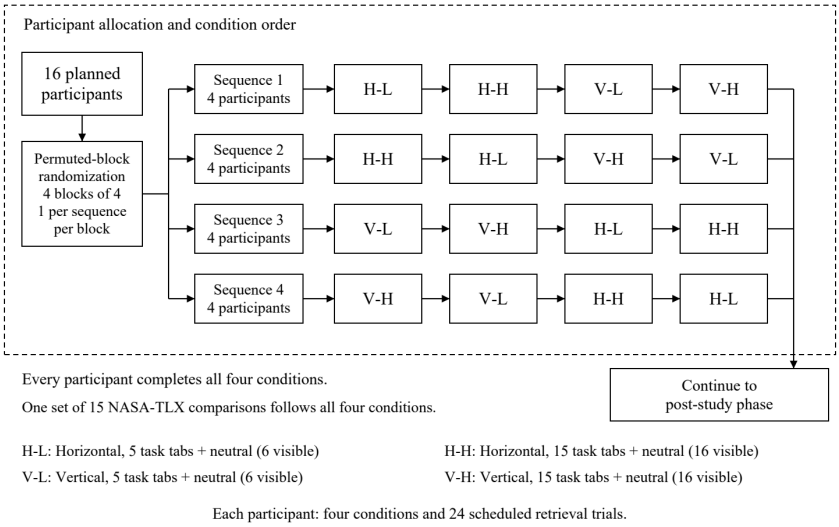


Figure 2: Participant allocation and counterbalancing. Sixteen participants will be assigned to four Latin-square sequences using permuted-block randomization with a block size of four. Each allocation block assigns one participant to each sequence, yielding four participants per sequence after four blocks. Every participant completes all four experimental conditions.

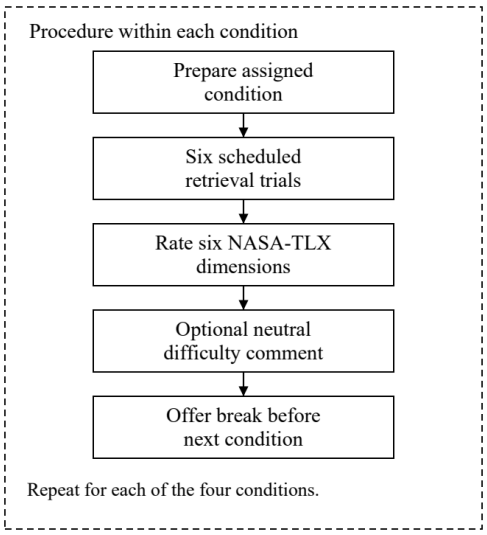


Figure 3: Procedure within each condition. Six scheduled trials are followed by NASA-TLX ratings, an optional neutral difficulty comment, and an offered break before the next condition.

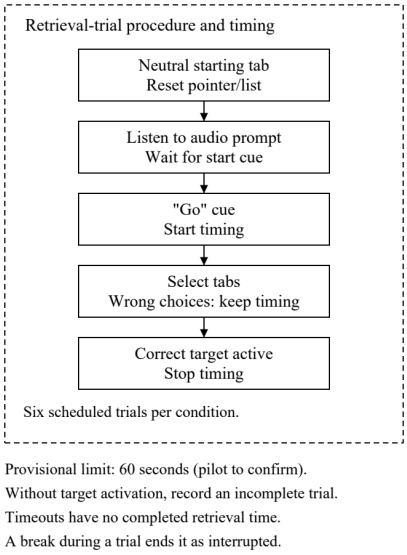


Figure 4: Retrieval-trial timing. Setup and prompt playback occur before timing starts at the Go cue. Correct target activation ends a completed trial; a timeout is recorded as incomplete.

perceived workload. The formal alternatives allow differences in either direction. RQ4 will be addressed through qualitative feedback. Incorrect-selection counts and total tab switches will remain additional descriptive measures of navigation activity.

4.3 Participants

The researchers plan to recruit 16 adult participants through their own personal contacts, classmates, and university networks using

convenience sampling. Participation will be voluntary, and no monetary compensation is planned. Participants may withdraw from the study at any time.

Participants must be at least 18 years old, regularly use a desktop or laptop browser, have prior experience working with multiple tabs, and usually keep those tabs within a single browser window. They must also be willing and able to attend an in-person session using the study setup and provide informed consent for the study procedures and screen recording. Prior experience with vertical tabs will not be required. A background questionnaire will record participants' usual browser, typical tab use, and prior experience with both orientations to provide context for interpreting their performance and feedback.

Initial eligibility will be assessed through a short self-report screening questionnaire before a session is scheduled. After providing informed consent, participants will receive the standardized briefing and practice, followed by a brief check of their ability to operate the required input devices and follow the retrieval-task instructions. Individuals who remain unable to perform these basic operations or understand the procedure after clarification will be excluded before the scored trials. Slow retrieval, ordinary practice errors, and unfamiliarity with vertical tabs will not, by themselves, be grounds for exclusion.

The planned sample size reflects the time and resources available for this course study and allows four participants to be assigned to each of the four counterbalanced orders. Every participant will complete all four experimental conditions. The eligibility criteria focus the study on users with relevant multi-tab browsing experience while reducing difficulties caused by basic device operation or misunderstanding of the task. However, the small convenience sample and the requirement for usual single-window browsing limit how broadly the findings can be generalized.

Eligible participants will be assigned to the four predetermined Latin-square sequence groups using permuted-block randomization. Assignment will occur in blocks of four, with each block containing each sequence group exactly once in a randomized order. Four blocks will therefore allocate the planned 16 participants evenly across the four sequence groups, resulting in four participants per sequence. This procedure preserves the intended counterbalancing while reducing the likelihood that sequence assignment is systematically associated with recruitment order.

4.4 Study Setup and Materials

Apparatus and Browser Configuration. Sessions will be conducted individually in a quiet indoor location with one facilitator present. All participants will use the same researcher-provided macOS laptop running Google Chrome and the same external mouse. The Chrome version, display resolution and scaling, browser-window dimensions, page zoom, pointer settings, and other relevant display settings will remain fixed across sessions.

A separate study browser profile will be prepared with unrelated extensions disabled and notifications suppressed. The experiment will use Chrome's horizontal and vertical tab layouts within a single browser window. The browser-window dimensions will remain fixed, and the vertical tab panel will remain expanded at a predetermined width throughout both vertical conditions. Hover previews will be disabled during scored trials. Participants will use the external mouse to select tabs, with tab-list scrolling permitted when necessary.

The mouse pointer will begin each scored trial at the same predetermined location within the webpage content area, away from both the horizontal tab bar and the vertical tab panel. The same absolute screen position will be used across all four conditions and finalized during pilot testing. Rearranging, opening, closing, pinning, or grouping tabs, using tab search, and using keyboard shortcuts for tab navigation will not be permitted during scored trials.

Content Sets and Tab Load. Four topic-based content sets, A–D, will be prepared. Each set will contain fifteen related webpages: five target webpages used in the retrieval tasks and ten additional competing webpages. The low-load version will contain the five target webpages, whereas the high-load version will contain the same five targets together with the ten competing webpages.

One standardized neutral start page will remain open as an additional tab in every condition. It will contain no task-relevant information, will not belong to any content set, and will never serve as a retrieval target. This provides the same starting state before every scored trial. Consequently, participants will see six total tabs in low-load conditions and sixteen total tabs in high-load conditions.

Each participant will encounter each content set only once. Across participants, the sets will be counterbalanced across experimental conditions so that the same materials can be evaluated under different orientations and tab loads without creating within-participant familiarity with the same content set.

Trial Schedule and Materials. Each condition will contain six scored retrieval trials. All five target webpages will appear at least once, while one target webpage will appear a second time. The identity and position of the repeated target will vary across content sets according to predetermined schedules rather than occurring consistently at the same point in the trial sequence. This preserves the same general task structure while reducing predictability.

The schedules follow the same balancing constraints across content sets: five unique targets are presented, exactly one target is repeated, and the identity and position of that repetition vary across sets. Table 3 presents the predetermined six-trial schedule for each content set.

The schedules are not intended to eliminate possible learning or repetition effects; instead, their structure is kept comparable across content sets and rotated across experimental conditions so that these effects are not systematically associated with a particular tab orientation or load level.

For a given content set, the same target webpages, six-trial target schedule, and retrieval prompts will be used regardless of whether that set is assigned to a low- or high-load condition. The high-load condition will therefore differ from the low-load condition primarily in the number of additional competing tabs rather than in the number, novelty, or repetition of target webpages.

Retrieval prompts will describe the information associated with the intended webpage without reproducing its exact tab title. Each prompt will be delivered through a pre-recorded audio file using standardized wording, voice, pacing, and playback volume. Each

Table 2: Fixed target-tab positions and counterbalanced content-set assignments across experimental conditions.

(a) Target positions by content set and load

Content set / load	P1	P2	P3	P4	P5
A – Sports / high	2	4	8	11	15
B – Animals / high	1	6	9	12	14
C – Nature / high	3	5	7	10	13
D – Science / high	2	6	8	12	15
All sets / low	1	2	3	4	5

(b) Content-set assignment by condition

Order group	H-L	H-H	V-L	V-H
1	A	B	C	D
2	D	C	B	A
3	B	A	D	C
4	C	D	A	B

Note. P1–P5 identify the five target webpages within each content set. Positions count task tabs only, from left to right for horizontal tabs and from top to bottom for vertical tabs. Study Start remains browser tab 1; add one to a task position to obtain its browser tab index. The same arrangement applies to either orientation, and tab order remains fixed throughout all six trials in a condition block. Low load retains the five targets in the same relative order as high load. Four participants per order group give each condition four exposures to every content set and its corresponding arrangement. Panel (b) specifies assignments to conditions; chronological condition orders remain as shown in Table 1.

Table 3: Predetermined six-trial target schedules for the four content sets. Each schedule contains five unique targets and one repeated target.

Set	T1	T2	T3	T4	T5	T6
A	P1	P2	P1	P3	P4	P5
B	P2	P3	P4	P2	P5	P1
C	P3	P4	P5	P1	P3	P2
D	P4	P5	P1	P2	P3	P4

recording will conclude with the cue "Go," which will mark the beginning of the scored retrieval interval. Participants will be instructed to listen to the complete prompt and not begin navigating until this cue is heard. Separate webpages and pre-recorded prompts will be used during familiarization so that experimental content is not exposed before the scored conditions.

The content sets and prompts will be reviewed for comparable title length, prompt complexity, terminology, favicon distinctiveness, and overall retrieval difficulty. The materials file will document webpage titles, favicons, URLs, tab order, trial targets, repeated-target position, neutral start-tab configuration, and prompt wording for each content set.

Before each condition block, the assigned webpages will be loaded and checked, and their tab order will remain fixed throughout that block. Each participant will encounter each content set once according to the assignments in Table 1.

OBS Studio will be used for continuous screen and audio recording during each session. Recordings will capture the browser display, mouse pointer, active-page changes, and playback of the pre-recorded auditory prompts, including the "Go" cue used to mark the start of each scored trial. A pilot recording on the study laptop will confirm that tab labels, pointer movements, active-page changes, audio cues, and playback timestamps are sufficiently clear for reliable manual coding. The OBS version, recording resolution, frame rate, and audio configuration will be documented and kept constant across sessions.

Supporting materials will include the screening and consent forms, background questionnaire, standardized facilitator script, counterbalancing schedule, NASA-TLX rating forms, 15 pairwise-comparison cards, weight-tally and scoring worksheets, a prompt for an optional neutral comment on overall difficulty after each condition, final interview guide, observation sheet, and coding worksheet. The coding worksheet will link participant and trial identifiers with start-cue timestamps, tab-activation sequences, target-activation timestamps, and the resulting performance measures.

4.5 Protocol

All facilitators will follow the same written instructions and trial schedule. A pilot will test the complete procedure, prompt presentation, and recording setup before the main study. It will also check the proposed 60-second trial limit, participants' understanding of the NASA-TLX ratings and weighting exercise, and total session duration. Any changes will be finalized before main-study data collection and applied consistently across all main-study sessions; pilot data will not be included in the main results.

4.5.1 Pre-study. Initial eligibility screening will occur before the study session. At the session, the facilitator will explain the study activities, screen and audio recording, voluntary participation, and the participant's right to withdraw before obtaining informed consent. Each participant will then receive a study code and complete the background questionnaire. The facilitator will explain the retrieval task and permitted navigation actions using a standardized script without disclosing any expected advantage of either tab orientation. Participants will be instructed to complete each retrieval task as quickly and accurately as possible and to avoid discussing their experience until prompted after the timed trials.

Before any scored trials, participants will complete familiarization using separate practice webpages and pre-recorded auditory prompts that will not appear in the experimental content sets. Participants will receive 90 seconds of familiarization with each tab orientation. Practice order will also be balanced within each Latin-square sequence group. Of the four participants assigned to each sequence, two will complete horizontal-tab familiarization first and two will complete vertical-tab familiarization first. Practice order will be predetermined in the allocation schedule and documented for each participant. During familiarization, the facilitator may clarify procedural instructions and will confirm that participants can operate the required controls and understand the retrieval task. No minimum retrieval speed or accuracy score will be required.

The facilitator will also explain the six NASA-TLX dimensions, rating anchors, and pairwise-comparison procedure. Participants

Table 4: Experimental conditions. Each condition includes one additional neutral starting tab that is not a retrieval target.

Code	Orientation	Task Tabs	Total Visible Tabs	Trials
H-L	Horizontal	5	6	6
H-H	Horizontal	15	16	6
V-L	Vertical	5	6	6
V-H	Vertical	15	16	6

will practice completing the workload ratings and one example pairwise comparison using the practice tasks. They will be instructed to select the dimension that contributed more to their perceived workload rather than simply selecting the dimension with the numerically higher rating. All familiarization and practice data will be excluded from analysis.

Before the scored conditions begin, the facilitator will verify the participant's assigned condition and content-set sequence and confirm that the study laptop, browser configuration, audio playback, and recording setup are functioning correctly. The recording will be checked to ensure that the browser tabs, mouse pointer, active-page changes, and pre-recorded "Go" cue can be clearly identified. Participants will then have an opportunity to ask procedural questions. The trial schedule will specify each prompt, correct target, target sequence, and other predetermined trial settings. The neutral starting tab will remain constant across all scored trials.

4.5.2 Experimental Phase. Every participant will complete all four experimental conditions listed in Table 4. Figure 2 shows participant allocation to the four Latin-square sequence groups. Figures 3 and 4 show the procedure within each condition and the timing of an individual retrieval trial, respectively. Table 1 specifies the condition order and content-set assignment for each sequence group. Four participants will be assigned to each group, and every participant will complete each experimental condition and content set once.

Before each condition block, the facilitator will prepare the assigned tab orientation, tab load, and content set according to the participant's counterbalancing sequence. The facilitator will verify the predetermined tab order, vertical-panel width where applicable, pointer and browser settings, audio playback, and screen-recording configuration before beginning the block. These setup actions will occur before the participant begins the scored trials.

Each condition block will contain six scored target-tab retrieval trials. All five target webpages in the assigned content set will serve as targets at least once during the block, while one webpage will appear as the target a second time. The repeated webpage and the position of its second appearance will follow the predetermined content-set schedules shown in Table 3.

The same target sequence assigned to a content set will be used whether that set appears under low or high tab load. Low- and high-load conditions will therefore contain the same five unique target webpages, the same repeated target, and the same six retrieval prompts. Both load conditions will additionally include the same neutral starting tab. High load will differ through the presence of ten additional related competing webpages. Participants will not be informed which target will repeat or at which point in the condition the repetition will occur.

Before each trial, the facilitator will activate the standardized neutral starting tab. The same neutral tab will be used for every trial and experimental condition and will never serve as a retrieval target. This provides a consistent initial browser state without giving any target webpage additional exposure before the retrieval prompt is presented.

The facilitator will also restore the predetermined tab-list scroll position where applicable and place the pointer at the standardized starting position. These actions will occur outside the measured retrieval interval. The neutral starting tab will not be included in the experimental count of five or fifteen task-related tabs; consequently, participants will see six total tabs under low load and sixteen under high load.

Once the browser is prepared, the pre-recorded auditory prompt assigned to that trial will be played. Participants will be instructed to listen to the complete prompt without beginning navigation. Each recording will conclude with the cue "Go," at which point the participant may begin locating the target tab. The "Go" cue will mark the beginning of the measured retrieval interval.

Retrieval timing will end when the correct target tab first becomes active. Incorrect selections will not stop the trial, and participants may continue selecting tabs until they activate the target or reach the trial time limit. No additional page interaction or verbal confirmation will be required after the correct target has been selected. During the measured interval, the facilitator will not provide hints, confirm incorrect choices, repeat or clarify the prompt, request explanations, or ask participants about workload. Tab activations and retrieval timestamps will later be coded from the screen and audio recording using the definitions established in the Preliminaries. Researcher setup and reset actions will not be included in the scored measures.

The proposed trial limit will be 60 seconds, subject to confirmation during pilot testing. A silent timer will allow the facilitator to enforce the limit without providing additional cues to the participant. A trial that reaches the limit without activation of the correct target will be marked as incomplete and will not receive a completed retrieval-time value. First-selection information and observed tab activations will still be retained where available, while a trial in which no tab is selected will be identified separately. Timeouts will remain part of the study record rather than being treated as technical failures.

If an external interruption, required researcher assistance, recording failure, or loss of the intended browser configuration occurs during a trial, the facilitator will stop the trial and document the cause. Affected or unverifiable measurements will be flagged as missing. A prompt that has already been presented will not be repeated as a replacement scored trial because the participant would have received prior exposure to its target. Unaffected trials from the same condition will remain usable.

Participants may request breaks between trials, and these breaks will be documented. If a participant requests a break during a timed trial, that trial will end and be recorded as interrupted. The interrupted trial will not be repeated as a replacement scored trial. After the break, the facilitator will continue with any remaining

scheduled trials in the same condition block. Once all six scheduled trials in that block have ended, participants will complete the six NASA-TLX ratings and may provide a brief neutral comment about the condition's overall difficulty. Detailed questions about navigation strategies and orientation preferences will be reserved for the final post-study interview to reduce the possibility that discussing strategies changes performance in later conditions. A standardized short break will be offered between condition blocks, after which the next condition will be prepared according to the assigned counterbalancing sequence.

4.5.3 Post-Study. After all four experimental conditions and their post-condition measures, participants will complete one set of 15 NASA-TLX pairwise comparisons. The facilitator will present every unique pair of the six dimensions once, in a separately randomized order for each participant. For each pair, the participant will choose the dimension that contributed more to workload across the completed target-tab retrieval tasks. The responses will determine one participant-specific set of weights, which will be used for all four conditions. This exercise will occur before the final comparative interview.

After completing the weighting exercise, participants will take part in a brief comparative interview. They will be asked which tab orientation they preferred under low and high tab loads and the reasons for their preferences. The interview will also explore the difficulties they experienced, whether their navigation strategies differed across conditions, and whether prior experience with either orientation affected their experience. The facilitator will follow a shared interview guide and document responses in written notes. Participants will also have an opportunity to provide additional comments. The session will conclude with a debriefing explaining the study's purpose, an opportunity for participants to ask questions, and thanks for their participation.

4.6 Data Collection

4.6.1 Data Recording and Checking. The researchers will manually review the screen recordings to obtain retrieval time, first-selection accuracy, incorrect-selection counts, and total tab switches using the definitions in the Preliminaries. A shared recording sheet will identify the participant code, experimental condition, content set, trial, prompt, target tab, and relevant video timestamps. Each trial will also be marked as completed, timed out, interrupted, or affected by a technical problem. The six workload ratings for each condition, the 15 pairwise choices, the derived six weights, and written feedback will be linked to the same participant code.

A second researcher will independently check approximately one quarter of the recorded trials across participants and conditions, together with any unclear cases. Differences in recorded times, selected tabs, or event counts will be resolved by reviewing the video. The researchers will also check for missing entries, incorrect trial labels, and recording or transcription mistakes. Workload checks will verify all six ratings for each condition, all 15 unique pairwise responses, and weights ranging from zero to five that sum to 15. Genuine slow responses and ordinary selection errors will remain in the data. The data-processing steps will be tested using clearly labeled dummy records kept separate from the actual study results.

A trial that reaches the time limit without reaching the target will be recorded as incomplete, with no completed retrieval time assigned. If no tab is selected, the first-selection result will be recorded as not observed. Technical problems and interruptions will be documented separately from these outcomes. Only affected or unverifiable measurements will be marked as missing, while other usable observations will be retained. Participant names and consent records will be kept separately from the coded analysis files.

4.7 Data Analysis

4.7.1 Data Preparation and Descriptive Statistics. The collected data will be organized according to the four experimental conditions: horizontal-low, horizontal-high, vertical-low, and vertical-high. For each participant, the researchers will calculate the mean target-tab retrieval time, first-selection accuracy, and weighted NASA Task Load Index (NASA-TLX) score for each condition. Incorrect-selection counts and total tab switches will also be summarized descriptively.

The researchers will report the mean and standard deviation for continuous measures, alongside the number of valid observations for each condition. First-selection accuracy will be reported as a percentage, while incomplete trials, timeouts, and missing observations will be documented separately. Prior experience and condition order will be retained as contextual variables.

4.7.2 Quantitative Analysis. The researchers will summarize each participant's results separately for the four experimental conditions. Mean retrieval time will be calculated from verified completed trials. First-selection accuracy will be calculated as the percentage of valid trials with an observable first selection in which that selection was correct. Perceived workload will be represented by the weighted NASA-TLX score calculated using the procedure in the Preliminaries, with the same participant-specific weights applied to all four conditions. If no usable observations are available for a condition, its corresponding score will be missing. An overall workload score will require all six condition ratings and a complete, valid set of 15 pairwise responses. If the weighting responses are incomplete, the participant's overall workload scores will be recorded as missing, while available individual ratings will be retained. The number of observations contributing to each score will be reported.

Tables and graphs will summarize orientation differences averaged across both tab loads for RQ1, load differences averaged across both orientations for RQ2, and orientation-by-load interactions for RQ3. The interaction hypothesis will be primary, while the orientation and tab-load hypotheses will be supporting. Analyses will account for repeated measurements from each participant. Retrieval time and weighted NASA-TLX scores will be treated as continuous measures, while trial-level first-selection results will retain their binary coding. Accuracy percentages will also be reported descriptively. The combined findings from RQ1–RQ3 will address the main research question, with qualitative findings from RQ4 providing context.

Completed and timed-out trial counts will accompany average retrieval times, and no-selection counts will accompany first-selection accuracy. These summaries will include the number of valid attempts used in each calculation. Completion patterns will

be considered alongside speed because averages based only on completed trials may hide difficulties in conditions with more timeouts. Incorrect-selection counts and total tab switches will provide additional descriptions of navigation activity, with completed and incomplete attempts reported separately.

Prior experience and condition order will provide context when interpreting the results. Missing observations and disruptions experienced by participants will be documented, including any limitations they create for comparisons between conditions. A non-significant result will not be interpreted as proving that the two orientations perform identically.

4.7.3 Test of Normality. The normality assumption will be examined separately for target-tab retrieval time and weighted NASA-TLX. For each outcome, the researchers will calculate each participant's orientation difference averaged across tab loads, tab-load difference averaged across orientations, and change in the orientation difference between low and high loads. Shapiro–Wilk tests and Q–Q plots will assess the distributions of these within-participant differences. The null hypothesis for each normality test is that the relevant differences follow a normal distribution, using a significance level of $\alpha = 0.05$.

These checks will inform the choice between the parametric and non-parametric analyses described below. A non-significant normality test will not establish that the assumption holds, particularly with the planned small sample. If retrieval times are strongly skewed, a log transformation may be considered and the checks repeated. Retrieval time and workload will be assessed separately, and any transformation and analysis route will be finalized before testing the research hypotheses.

4.7.4 Comparison of Experimental Conditions. The analysis will examine the main effect of tab orientation (RQ1), the main effect of tab load (RQ2), and the orientation × load interaction (RQ3) on target-tab retrieval time, first-selection accuracy, and weighted NASA-TLX. Because the study's primary hypothesis concerns whether the difference between horizontal and vertical orientations changes with tab load, the interaction effect will be treated as the primary effect of interest; the two main effects will be treated as supporting.

Retrieval time and weighted NASA-TLX will each be analyzed as a single continuous outcome with two within-subjects factors, orientation and tab load, each at two levels. For an outcome classified as parametric, a two-way (multifactorial) repeated-measures ANOVA will be used. For an outcome classified as non-parametric, an aligned rank transform repeated-measures ANOVA will be used instead. Either analysis tests the orientation main effect ($H_{0,1}$), the tab-load main effect ($H_{0,2}$), and the interaction ($H_{0,3}$), and partial η^2 will be reported as the effect size. For an outcome on the parametric route, the hypotheses therefore concern differences in condition means. For an outcome on the non-parametric route, they concern effects on aligned ranks, that is, shifts in the distribution of values, and significant results will be interpreted in that way rather than as differences in means. Because each factor has only two levels, every effect has one degree of freedom, and a sphericity correction will not be needed.

If the interaction is significant for an outcome, the pattern will be compared with the expected pattern, in which the vertical advantage grows with tab load, and followed by pairwise interaction comparisons (differences of differences). In this 2×2 design, these comparisons are the orientation difference at low load and the orientation difference at high load. Each will use a paired t-test for a parametric outcome or a Wilcoxon signed-rank test for a non-parametric outcome, with Cohen's d or the rank-biserial correlation r as the effect size. A significant main effect will not need a post hoc test, because each factor has only two levels.

First-selection accuracy is recorded as correct or incorrect on each trial, so the Shapiro–Wilk test alone does not determine its analysis. The probability of a correct first selection will be modeled at the trial level with a logistic model containing orientation, tab load, their interaction, and a random intercept for each participant to account for repeated trials. Each effect will be tested with a likelihood ratio test comparing nested models. The interaction ($H_{0,3}$) is tested by comparing the full model with the model that omits the interaction term. Each main effect ($H_{0,1}$ and $H_{0,2}$) is tested by comparing the model containing both factors without the interaction with the model that also omits that factor. Trials with no observable first selection are excluded, as defined in the Preliminaries. If this model cannot be fitted, each participant's first-selection accuracy percentage per condition will instead be treated as a continuous outcome and analyzed through the same route as retrieval time and workload; the hypotheses for accuracy would then concern mean accuracy percentages rather than the probability of a correct selection on a single trial. Incorrect-selection counts and total tab switches will serve as additional descriptive measures.

Each joint hypothesis pair ($H_{0,1}$–$H_{1,1}$, $H_{0,2}$–$H_{1,2}$, and $H_{0,3}$–$H_{1,3}$) will be evaluated through its three outcome-specific tests, one for retrieval time, one for first-selection accuracy, and one for weighted NASA-TLX. Because each null hypothesis is rejected if any one of its three outcomes shows an effect, the three p-values for a given effect will be adjusted using the Holm procedure. A null hypothesis will be rejected if at least one of its three adjusted p-values is at or below 0.05, and the outcome-level results will be reported to show which outcomes account for the rejection.

All statistical tests will use a significance level of 0.05. The results will be interpreted in relation to the research questions and the formal hypotheses stated in Section 4.2, with effect sizes and uncertainty reported where appropriate. A non-significant result will not be taken as evidence of equivalence. Qualitative feedback (RQ4) will be used to contextualize the numerical findings.

4.7.5 Qualitative Analysis. Two researchers will review the written responses collected after each condition and during the final interview. They will identify recurring comments about navigation difficulties, strategies, workload, and orientation preferences, compare their interpretations, and group related comments while retaining differing experiences. The responses will remain linked to their orientation and tab-load conditions to address RQ4.

These comments will be considered alongside the numerical results to help explain why participants found particular conditions easier or more difficult. Written summaries will be presented as paraphrases, and only wording documented exactly will be used as a direct quotation.

4.7.6 Ethical Considerations. The study will prioritize the rights, privacy, and well-being of all participants throughout the research

process. Before participating, individuals will receive a clear explanation of the study's purpose, procedures, expected activities, and data collection methods. Informed consent will be obtained before any study activities or recordings begin. Participants will be informed that participation is voluntary, that they may decline to answer optional questions, and that they may withdraw from the study at any time without penalty.

To protect participant privacy, all experimental sessions will use a researcher-provided laptop and a separate study browser profile containing prepared webpages. Participants will not be required to log in to personal accounts or disclose passwords, private messages, or other personal information. Screen and audio recordings will be limited to the experimental session and will be used to measure target-tab retrieval time, first-selection accuracy, incorrect-selection counts, and total tab switches. Recordings will also support the verification of trial measurements and the interpretation of participants' feedback.

Each participant will be assigned a study code to maintain confidentiality during data processing and analysis. Participant names and consent records will be stored separately from coded research data. Access to identifiable information and raw recordings will be restricted to the researchers who need them for the study. Research findings will be presented in aggregated form or through anonymized summaries of participant feedback. Any direct quotations will only be used when the wording has been documented accurately and the quotation can be presented without revealing the participant's identity.

Participants will not be evaluated based on their individual browsing speed or accuracy, and ordinary navigation errors or unfamiliarity with vertical or horizontal tabs will not be treated as grounds for exclusion. They may request breaks between trials, and any interruptions or technical difficulties will be documented according to the study protocol. The researchers will handle collected data confidentially and use them only for the stated research and academic purposes. Data storage, access, and retention will follow the applicable university requirements and the procedures communicated to participants during informed consent.

5 Future Work

The immediate next step is to finalize the webpage sets and retrieval prompts and conduct a pilot of the proposed experiment. The pilot will check whether the instructions are clear, the familiarization period is sufficient, and the provisional 60-second trial limit is appropriate. It will also check screen and audio recording quality, participants' understanding of the NASA-TLX ratings and weighting exercise, the time required to complete these measures, expected session duration, and whether the study setup and settings can be maintained consistently across sessions on the same macOS laptop. These observations will guide revisions to the materials and procedure before the main experiment. The researchers will also confirm the statistical analysis plan with the professor and test the data-processing steps using clearly labeled dummy data before collecting the main study data.

Software-assisted collection of tab activations and timestamps may also be explored to reduce manual coding effort. Any proposed tool will first be checked against the screen recordings to determine whether its records match the study's trial boundaries and measurement definitions. Screen recording and manual coding will remain the planned collection method unless an alternative is verified. Once the materials, procedure, and analysis plan are finalized, participants will be recruited for the main experiment.

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